
Assignment # 4 [CLO 3]

INSTRUCTIONS:

- Max Number of Students in a Group: 3
- Deadline: 16th Dec 2025
- Marks will be deducted for late submission

ATELIER B

Download **atelierb** from the link given below:

<https://www.atelierb.eu/en/atelier-b-support-maintenance/download-atelier-b/>

Atelier B

- Atelier B is a widely used industrial-grade environment for the B Method. It supports writing B specifications (machines/refinements), generating proof obligations, and performing proofs, like B-Toolkit.
- It provides a GUI (using Qt) or a command-line interface, so it works on **Windows, Linux, macOS**.
- It supports — besides modelling and specification — things like project management, proof support, refinements, and even translation of B specifications to implementation code

QUESTION # 1

Step-by-Step Atelier B Usage (easy guide):

Assignment # 4 [CLO 3]

Step 1: Open Atelier B

Start Menu → **Atelier B Community Edition** → **BToolkit/BPlatform**

Step 2: Create a New Project

- Click **File** → **New Project**
- Enter project name
- Select folder
- Click **OK**

This creates a workspace for your B machines.

Step 3: Add a New Abstract Machine

- Right-click **Machines**
- Choose **New**
- Name it: *BankAccount*
- A blank editing window opens

Paste the sample machine here.

Step 5: Type Checking

Once parsing is OK, click:

Analyse → **Type Check**

It verifies:

- Variable types
- Expressions
- Sets
- Guards

All good → green ✓

Step 6: Generate Proof Obligations (POs)

Click:

Prove → **Generate PO**

Atelier B will create several proof obligations:

- Invariant initialization proof
- Invariant preservation (for each operation)
- Precondition correctness

You will now see POs listed on the left panel.

Assignment # 4 [CLO 3]

Step 7: Automatic Provers

Click:

Prove → Automatic Proof

Many POs will turn **green** automatically.

Unproven ones remain **red**.

Step 8: Manual/Interactive Prover (optional)

For remaining POs:

- Right-click PO
- Choose **Interactive Prover**

You can:

- Expand definitions
- Simplify
- Apply arithmetic rules

Once proof is completed → ✓ green

QUESTION # 2

Design an Abstract Machine for a Library System.

Scenario:

A library system records books and who borrowed them.

Tasks:

1. Create an abstract machine named LibrarySystem
2. Variables:
 - books (set of available books)
 - issued (mapping of book → student)

Assignment # 4 [CLO 3]

3. Invariant examples:

- $\text{issued} \subseteq \text{books} \times \text{STUDENTS}$

4. Operations:

- $\text{addBook}(b)$
- $\text{issueBook}(b, s)$
- $\text{returnBook}(b)$

You must:

- Write the abstract machine
- Parse and type-check
- Generate proof obligations
- Prove at least 2 POs